Week #**8:** [Neurovascular Supply of the Proximal Upper Extremity]

(Hyperlinked = anatomy notes below)

• Introduction to Module

Vascular Supply to the Proximal Upper Extremity

- **Vascular Supply (11 min)**

Nerve Supply to the Proximal Upper Extremity

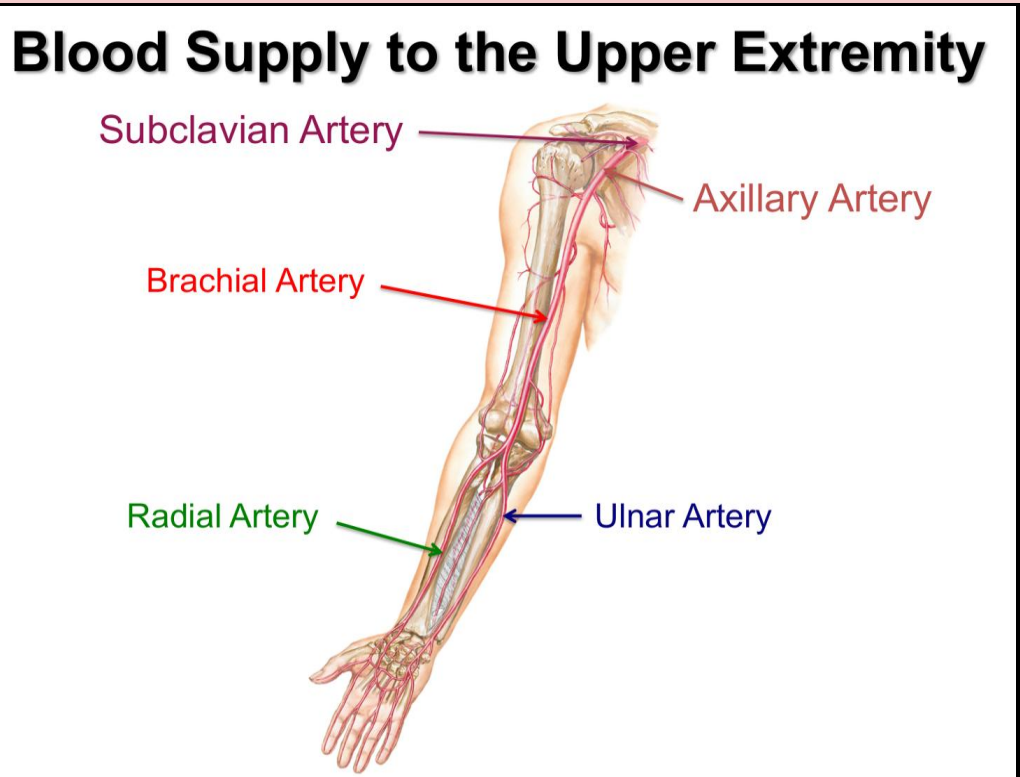
- **Nerve Supply (18 min)**

What have you learned FROM ALL PROXIMAL UPPER EXTREMITY MODULES?

- Clinical Approach to the Upper Extremity Quiz (10 min)
- Evaluate this Module

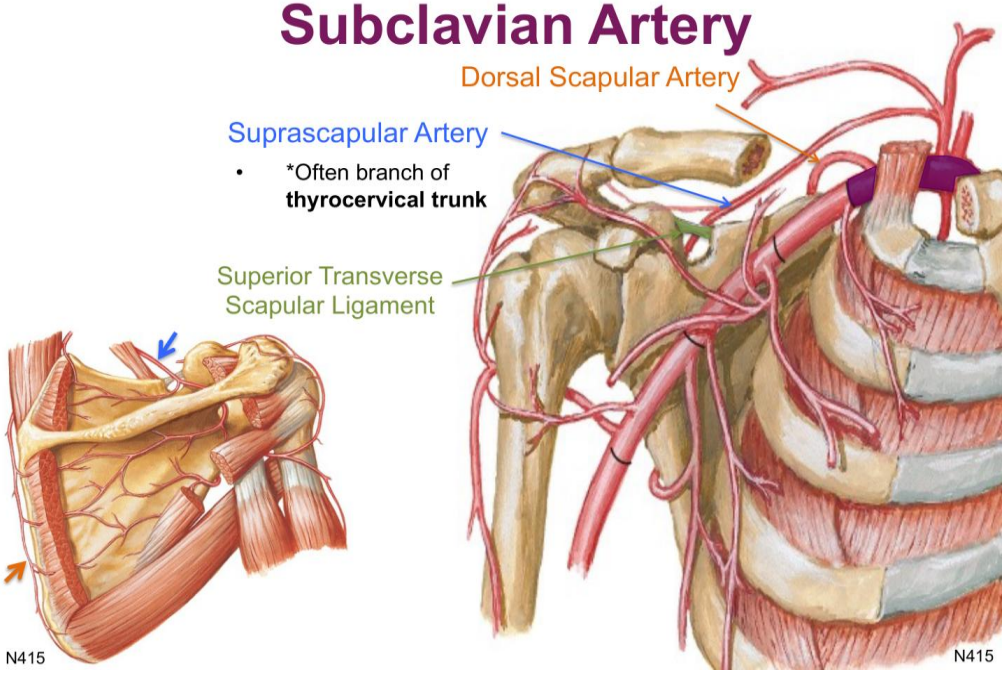
Vascular Supply

- Vascular Supply of the Proximal Upper Extremity



Notes

Vascular Supply to the Axilla:
Subclavian Artery



B. Axillary Artery – at the lateral border of the 1st rib, the subclavian artery changes names to the axillary artery. The artery courses through the axilla and will change names again at the inferior border of teres major to the **brachial artery**. The axillary artery can be subdivided into three parts, by pectoralis major which lies anterior to it. The axillary artery gives off many branches. Branches of note contribute to an extensive vascular anastomoses around the scapula.

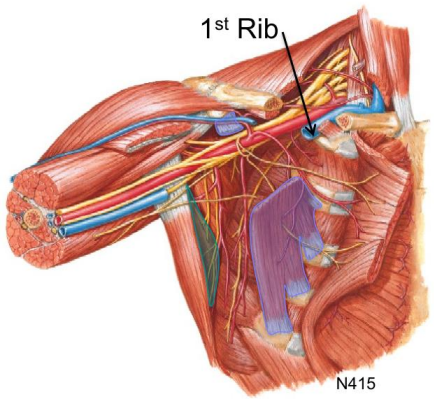
1. 1st Part of Axillary Artery – from the lateral border of the 1st rib to the superior border of pectoralis minor.
   a. *For your interest only:* It gives off one artery - Superior Thoracic Artery – small artery that supplies the upper thoracic wall
2. 2nd Part of Axillary Artery – deep (posterior) to pectoralis minor. Gives off two branches:
   a. **Thoracoacromial Artery** – branches supply the acromial, pectoral, clavicular, and deltoid regions contributing to the glenohumeral anastomosis
   b. *For your interest only:* Lateral Thoracic Artery – supplies the lateral thoracic wall, pectoral region, and breast; runs with long thoracic nerve
3. 3rd Part of Axillary Artery – from inferior border of pectoralis minor to inferior border of teres major. Gives off three arteries:
   a. **Subscapular Artery -** a short trunk off the 3rd part of the axillary artery that divides into the thoracodorsal artery and the circumflex scapular artery
      i. **Circumflex scapular artery** – curves around the lateral border of the scapula, coursing deep to teres minor and infraspinatus to contribute to the scapular anastomosis
      ii. Thoracodorsal Artery – courses with the thoracodorsal nerve to supply the superficial back
   b. **Anterior Humeral Circumflex Artery –** encircles the surgical neck of humerus laterally, anastomoses with posterior humeral circumflex artery and gives off a branch to the glenohumeral joint
   c. **Posterior Humeral Circumflex Artery –** encircles the surgical neck of the humerus medially, anastomoses with the anterior posterior humeral circumflex artery, and continues to run with the axillary nerve to contribute to the glenohumeral anstamosis.

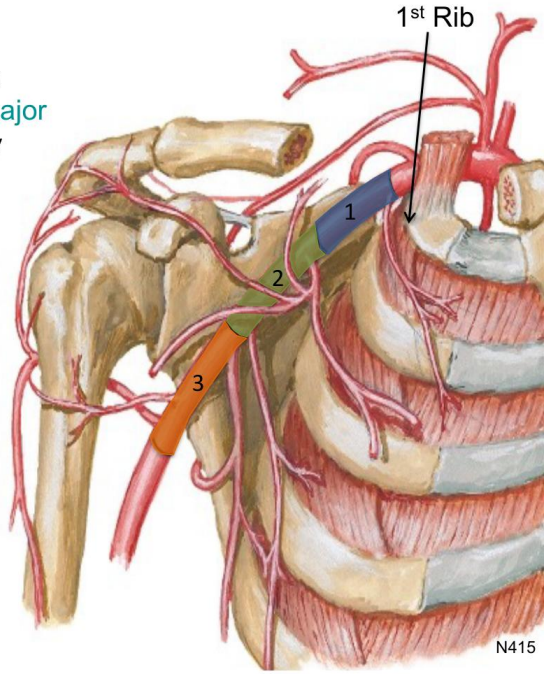
C. Brachial Artery – continuation of the **axillary artery** distal to the inferior margin of teres major. Travels through the arm medial to the biceps brachii and anterior to coracobrachialis and brachialis. Brachial artery terminates in the cubital fossa as the **radial artery** laterally and the **ulnar artery** medially. During its course it gives off 3 major branches in the arm:

1. **Profunda Brachii Artery** – which will course to the posterior compartment to supply it. Nearing the elbow the profunda brachii artery will change names to the **radial collateral artery**, which contributes to the collateral circulation surrounding the elbow.

Vascular Supply to the Axilla

- From lateral border of 1st rib
- To Inferior border of teres major
- Subdivided in three parts by pectoralis minor

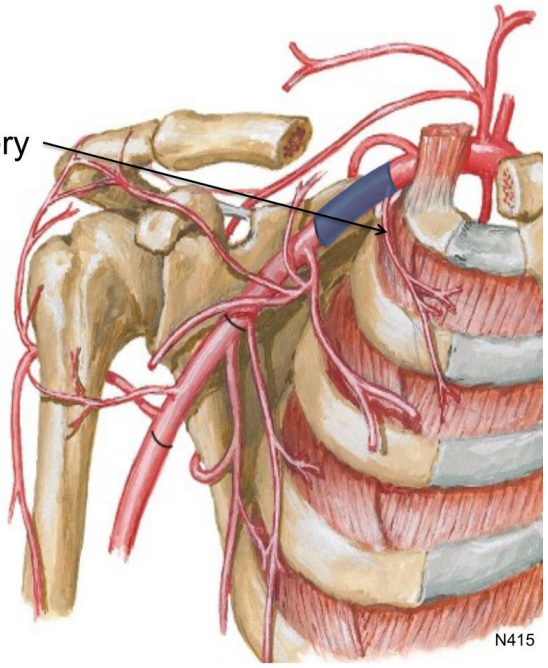




I. Vasculature of the Proximal Upper Extremity:

a. **Subclavian Artery** – branches from the brachiocephalic trunk on the right and directly off the arch of the aorta on the left. Gives off many branches to supply the head, neck, and thorax. Two branches of the subclavian artery contribute to the vascular anastomoses around the scapula:
   i. ***Dorsal Scapular Artery*** – heads posteriorly to supply the superior superficial back musculature
   ii. ***Suprascapular Artery*** – branch of the subclavian artery (or thyrocervical trunk) that passes posteriorly with the suprascapular nerve; runs superior to the transverse scapular ligament to supply the posterior shoulder

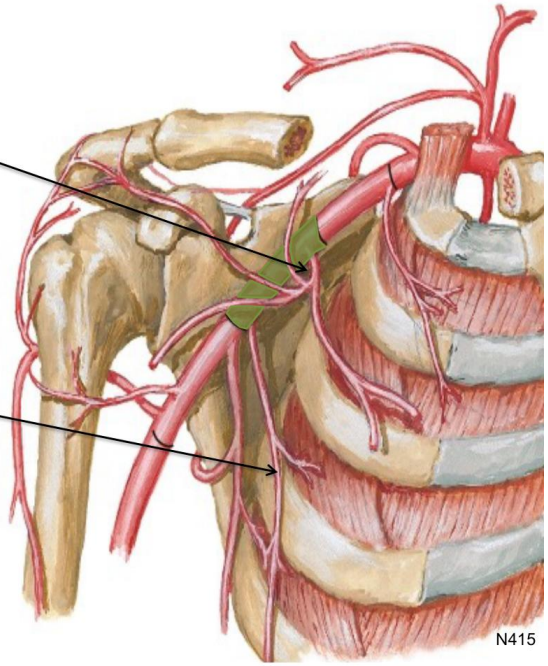
Axillary Artery: 1st Part



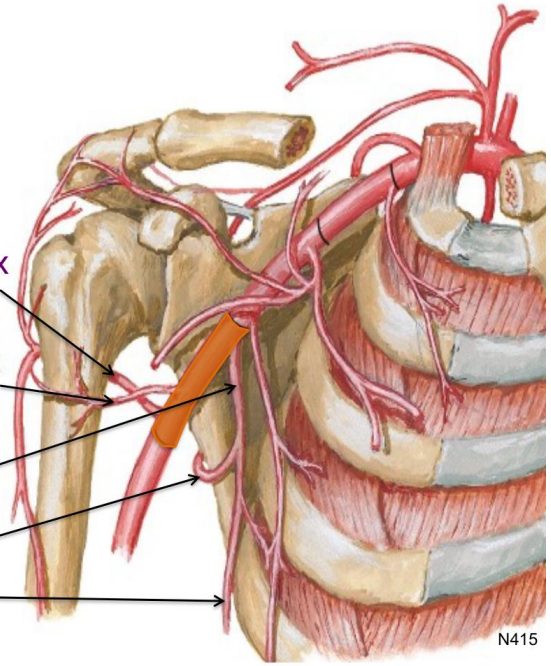
Axillary Artery: 2nd Part

Thoracoacromial Artery
- Acromion branch
- Clavicular branch
- Deltoid branch
- Pectoral branch

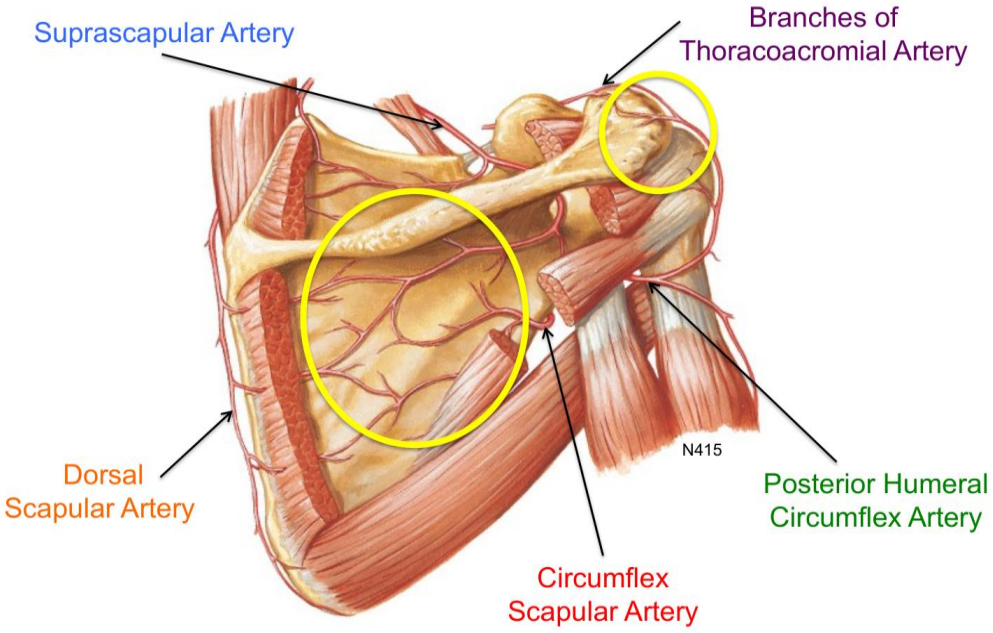
Lateral thoracic artery
- Runs with long thoracic nerve



Axillary Artery: 3rd Part

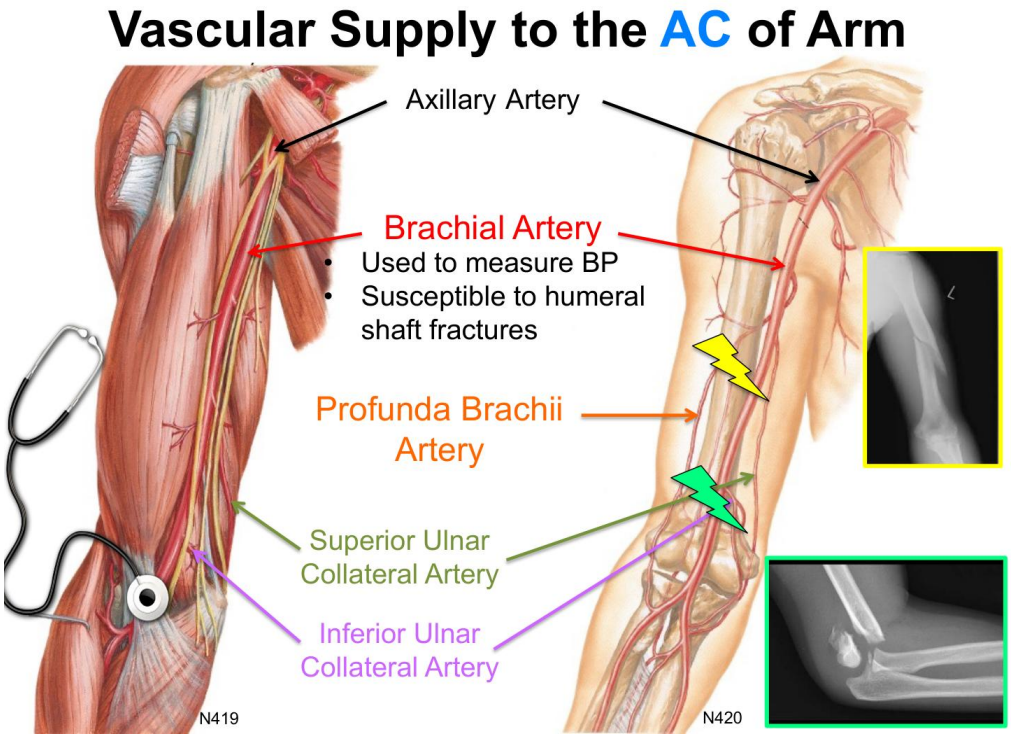


Vascular Anastomoses



B. Axillary Artery – at the lateral border of the 1st rib, the subclavian artery changes names to the axillary artery. The artery courses through the axilla and will change names again at the inferior border of teres major to the **brachial artery.** The axillary artery can be subdivided into three parts, by pectoralis major which lies anterior to it. The axillary artery gives off many branches. Branches of note contribute to an extensive vascular anastomoses around the scapula.

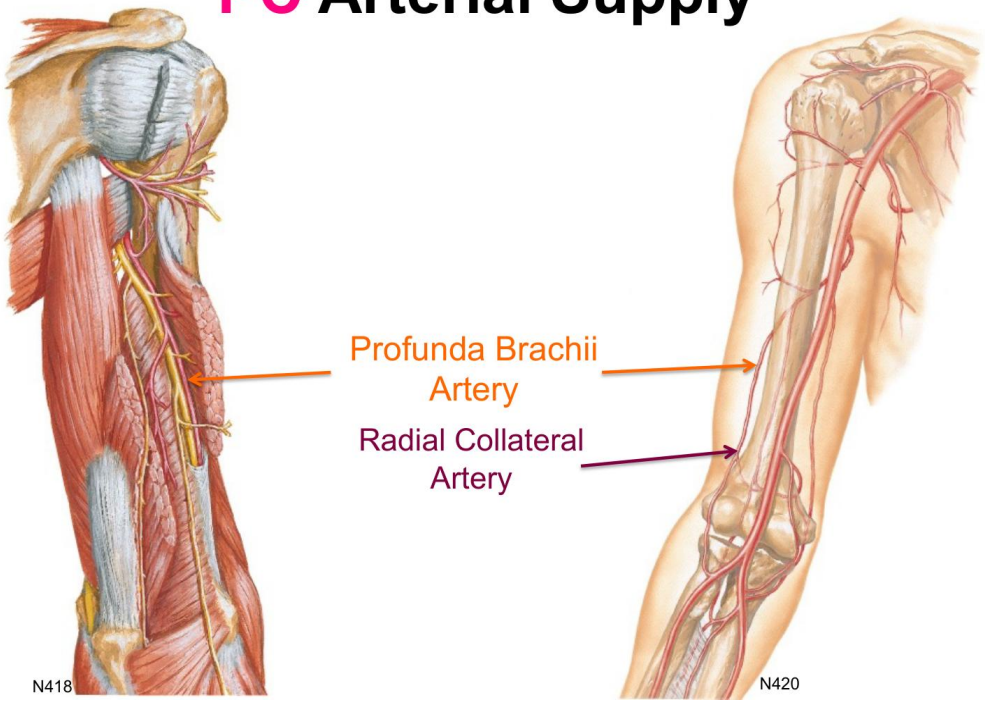
1. **1st Part of Axillary Artery** – from the lateral border of the 1st rib to the superior border of pectoralis minor.
    a. For your interest only: It gives off one artery - **Superior Thoracic Artery** – small artery that supplies the upper thoracic wall
2. **2nd Part of Axillary Artery** – deep (posterior) to pectoralis minor. Gives off two branches:
    a. **Thoracoacromial Artery** – branches supply the acromial, pectoral, clavicular, and deltoid regions contributing to the glenohumeral anastomosis
    b. *For your interest only*: **Lateral Thoracic Artery** – supplies the lateral thoracic wall, pectoral region, and breast; runs with ***long thoracic nerve***
3. **3rd Part of Axillary Artery** – from inferior border of pectoralis minor to inferior border of teres major. Gives off three arteries:
    a. **Subscapular Artery** - a short trunk off the 3rd part of the axillary artery that divides into the thoracodorsal artery and the circumflex scapular artery
        i. ***Circumflex scapular artery*** – curves around the lateral border of the scapula, coursing deep to teres minor and infraspinatus to contribute to the scapular anastomosis
        ii. ***Thoracodorsal Artery*** – courses with the thoracodorsal nerve to supply the superficial back
    b. **Anterior Humeral Circumflex Artery** – encircles the surgical neck of humerus laterally, anastomoses with posterior humeral circumflex artery and gives off a branch to the glenohumeral joint
    c. **Posterior Humeral Circumflex Artery** – encircles the surgical neck of the humerus medially, anastomoses with the anterior posterior humeral circumflex artery, and continues to run with the axillary nerve to contribute to the glenohumeral anastomosis



C. Brachial Artery – continuation of the **axillary artery** distal to the inferior margin of teres major. Travels through the arm medial to the biceps brachii and anterior to coracobrachialis and brachialis. Brachial artery terminates in the cubital fossa as the **radial artery** laterally and the **ulnar artery** medially. During its course it gives off 3 major branches in the arm:

1. **Profunda Brachii Artery** – which will course to the posterior compartment to supply it. Nearing the elbow the profunda brachii artery will change names to the radial collateral artery, which contributes to the collateral circulation surrounding the elbow.
2. **Superior Ulnar Collateral Artery** – which will contribute to a network of collateral circulation surrounding the elbow.
3. **Inferior Ulnar Collateral Artery** – which will also contribute to a network of collateral circulation surrounding the elbow.

PC Arterial Supply

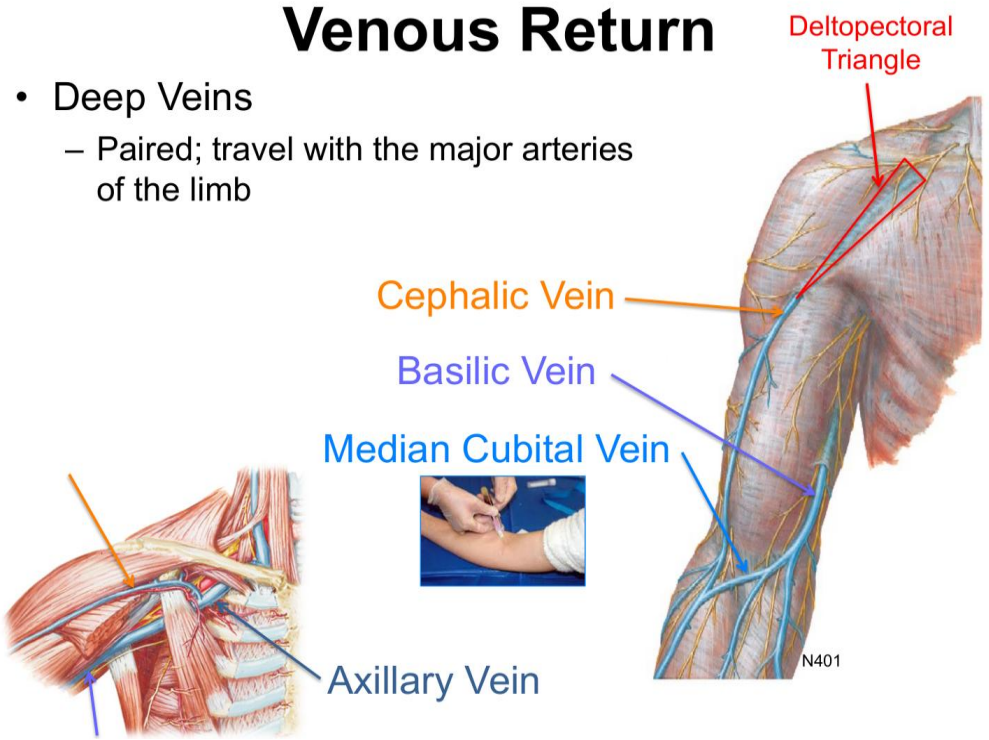


Clinical Note: To measure blood pressure the pressure cuff is placed around the arm and inflated until it compresses the brachial artery. A stethoscope is placed over the brachial artery located in the cubital fossa to listen to the sounds of the blood coursing through the artery.

Clinical Note: The brachial artery lies adjacent to the medial aspect of the humeral shaft and thus is susceptible to injury if there is a fracture to the mid-shaft or the supracondylar areas of the humerus.
The profunda brachii artery runs closely adhered to the humerus in the radial groove and thus is also susceptible to mid-shaft fractures.

Venous Return

- Deep Veins
  - Paired; travel with the major arteries of the limb

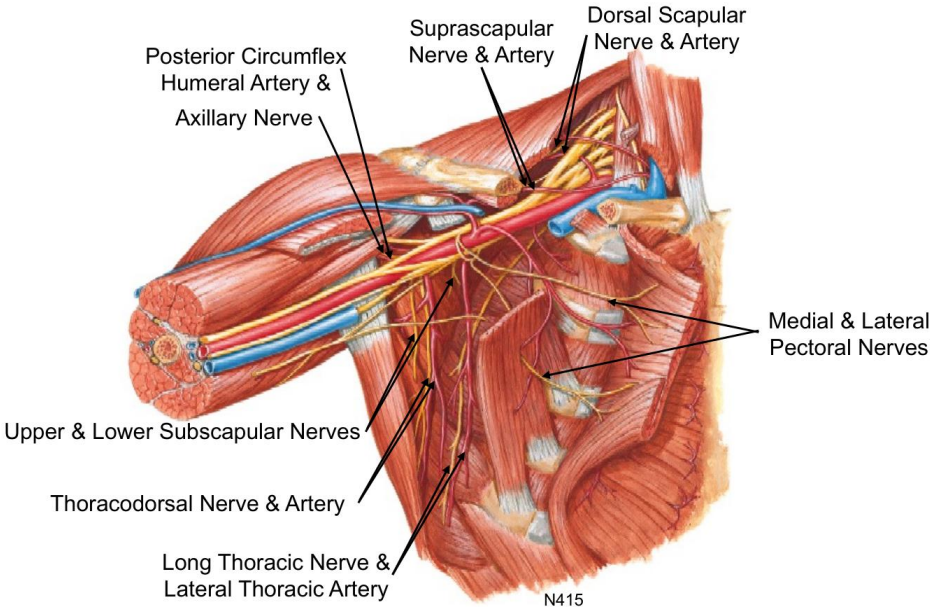


Venous Return of the Upper Limb
1. **Cephalic vein** – ascends on the anterolateral surface of the forearm and arm; drains into the terminal axillary vein
2. **Basilic vein** – ascends along the medial side of the forearm and inferior part of the arm; forms the axillary vein with the brachial veins
3. **Median Cubital vein** – passes obliquely in the cubital fossa connecting the cephalic and basilica veins

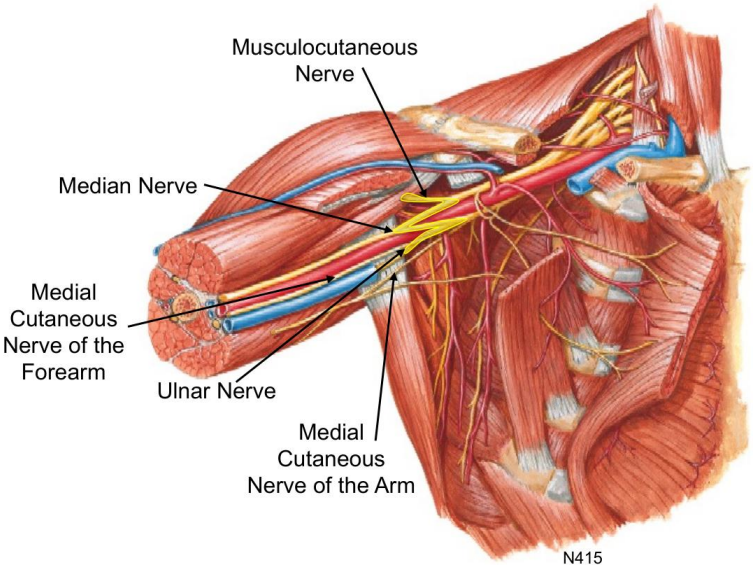
Clinical Note: the median cubital vein is often used to draw blood and for intravenous access.

Nerve Supply

Neurovascular Supply to the Pectoral Girdle



Neurovascular Supply to the Arm



Brachial Plexus:

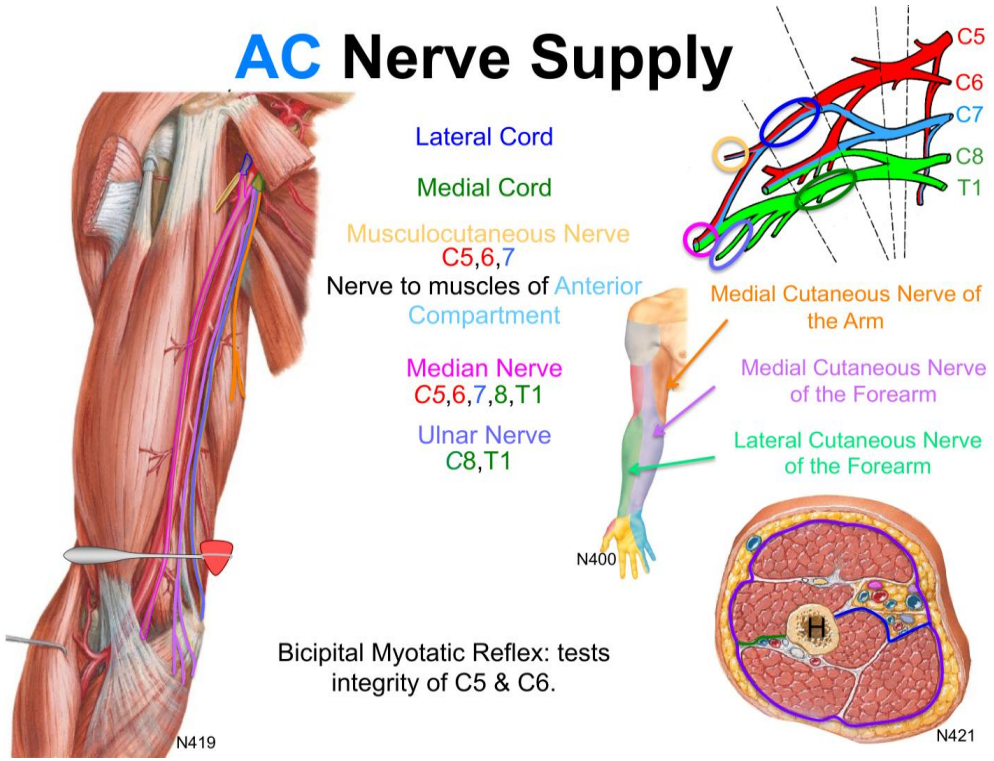
Roots	Branches
C5, C6, C7, C8, T1	Dorsal scapular nerve: arises from C5 root; pierces middle scalene muscle before coursing posteriorly to innervate the rhomboid muscles.
	Long thoracic nerve: arises from roots of C5-C7; motor to serratus anterior muscle along its superficial aspect
Trunks	**Branches**
Superior: union of C5 & C6	Suprascapular nerve: provides motor innervation to supraspinatus & infraspinatus muscle
Middle: C7 root	No nerves
Inferior: union of C8 & T1	No nerves
Divisions	Each trunk divides into an anterior and posterior division (6 total divisions)
Anterior	No Branches - Nerves arising from the anterior divisions generally supply the **flexor** (anterior) compartments of the upper extremity.
Posterior	No Branches - Nerves arising from the posterior divisions generally supply the **extensor** (posterior) compartments of the upper extremity.
Cords	**Branches**
Posterior: union of three posterior divisions	Upper subscapular nerve: motor to subscapularis
	Thoracodorsal nerve (aka middle subscapular nerve): motor to latissimus dorsi
	Lower subscapular nerve: motor to subscapularis and teres major
	Axillary nerve: motor to teres minor & deltoid; sensory to skin of shoulder
	Radial nerve: motor to extensors of posterior arm & forearm; sensory along posterior aspect of upper extremity.
	Lateral pectoral nerve: motor to pectoralis major & minor; travels superomedial to pectoralis minor

B. Nervous Structures in Arm:

1. **Musculocutaneous Nerve (C5, C6, C7) –** supplies all of the muscles in the anterior compartment of the arm. Arises from the lateral cord of the brachial plexus, it then pierces the coracobrachialis and continues distally between the brachialis and the biceps. After supplying all the muscles in the anterior compartment it will give off the **lateral cutaneous nerve of the forearm,** which supplies cutaneous innervation to the lateral forearm.
   - ❖ ***Clinical Note:*** The bicipital myotatic reflex test assesses the integrity of C5 and C6 spinal levels.
2. **Median Nerve** (C5**, C6, C7, C8, T1**)– formed by the union of the medial and lateral roots from the medial and lateral cords of the brachial plexus. Initially it courses lateral to the brachial artery until it reaches the middle of the arm where it crosses medially to the brachial artery. It then descends deep to the bicipital aponeurosis in the cubital fossa where it lies deep to the median cubital vein.
3. **Ulnar Nerve** (C7**, C8, T1**) – arises from the medial cord of the brachial plexus. Travels medially in the anterior compartment until approximately mid-shaft when it pierces the medial intermuscular septum to travel in the posterior compartment.
4. **Medial Cutaneous Nerve of the Arm (medial brachial cutaneous nerve)** – arises from the medial cord of the brachial plexus. Courses adjacent to the ulnar nerve and branches into the superficial fascia of the medial arm to supply the medial arm and axilla with cutaneous innervation.
5. **Medial Cutaneous Nerve of the Forearm (medial antebrachial cutaneous nerve)** – arises from the medial cord of the brachial plexus. Courses adjacent to the ulnar nerve to the superficial fascia of the forearm where it will branch to supply the anteromedial aspect of the forearm with cutaneous innervation.
6. **Radial Nerve (C5, C6, C7, C8,** T1**) –** The radial nerve is one of the terminal branches from the posterior cord. It enters the arm posterior to the brachial artery, medial to the humerus and anterior to the long head of the triceps. It descends inferolaterally with the profunda brachii artery in the radial groove. Supplies the triceps brachii and then pierces the lateral intermuscular septum and continues anteriorly between the brachialis and the brachioradialis to reach the forearm.
   - ❖***Clinical Note:*** Due to the radial nerves close association with the humeral shaft as it runs within the radial groove it is susceptible to injury during humeral mid-shaft fractures. The radial nerve courses with the profunda brachii artery and thus it is also at risk of injury.

B. Cutaneous Innervation of the Upper Limb

1. **Posterior cutaneous nerve of the arm and forearm** (C5-C8) – branch of the radial nerve; supplies the skin on the posterior arm and forearm

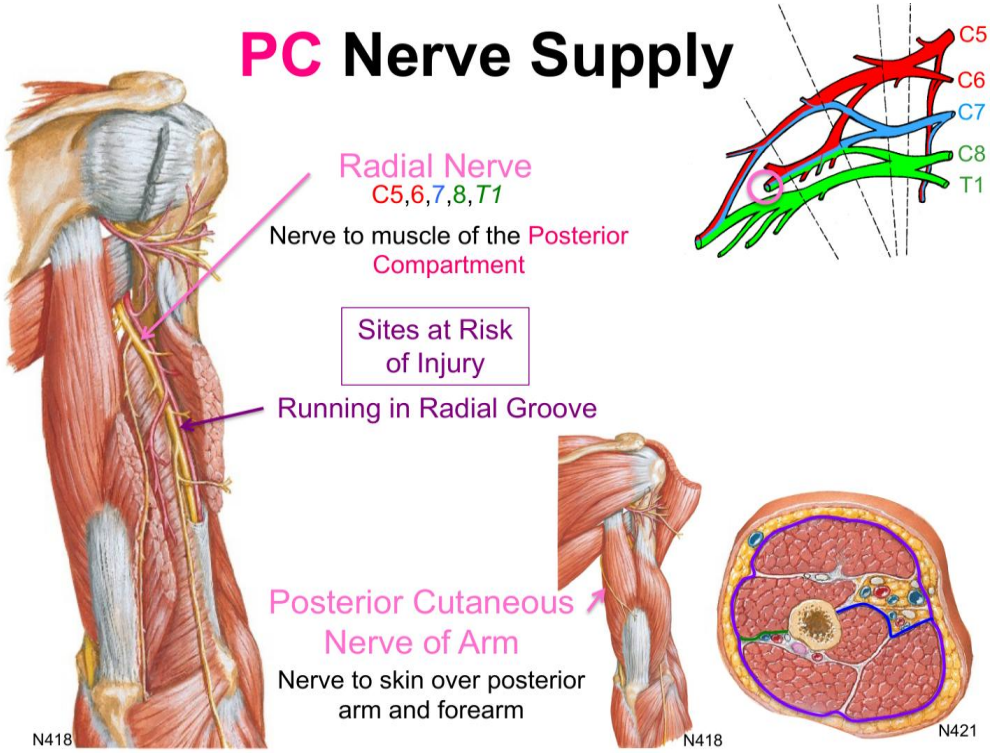


B. Nervous Structures in Arm:

1. **Musculocutaneous Nerve** (C5, C6, C7) – supplies all of the muscles in the anterior compartment of the arm. Arises from the lateral cord of the brachial plexus, it then pierces the coracobrachialis and continues distally between the brachialis and the biceps. After supplying all the muscles in the anterior compartment it will give off the **lateral cutaneous nerve of the forearm**, which supplies cutaneous innervation to the lateral forearm.

Clinical Note: The bicipital myotatic reflex test assesses the integrity of C5 and C6 spinal levels.

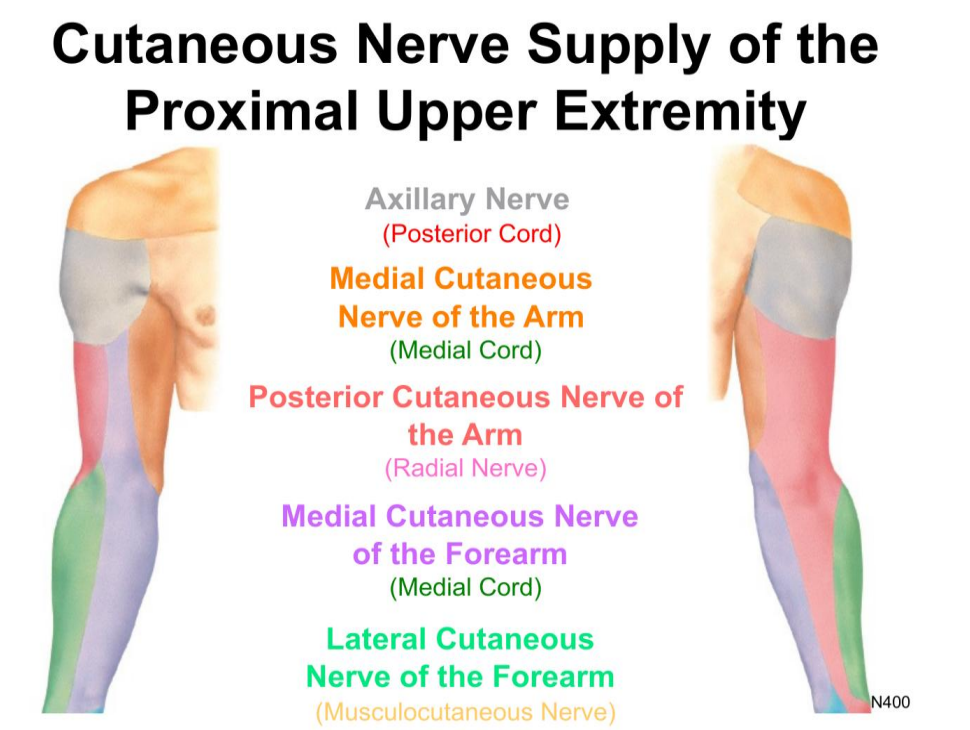
2. **Median Nerve** (C5, C6, C7, C8, T1)– formed by the union of the medial and lateral roots from the medial and lateral cords of the brachial plexus. Initially it courses lateral to the brachial artery until it reaches the middle of the arm where it crosses medially to the brachial artery. It then descends deep to the bicipital aponeurosis in the cubital fossa where it lies deep to the median cubital vein.
3. **Ulnar Nerve** (C7, C8, T1) – arises from the medial cord of the brachial plexus. Travels medially in the anterior compartment until approximately mid-shaft when it pierces the medial intermuscular septum to travel in the posterior compartment.
4. **Medial Cutaneous Nerve of the Arm (medial brachial cutaneous nerve)** – arises from the medial cord of the brachial plexus. Courses adjacent to the ulnar nerve and branches into the superficial fascia of the medial arm to supply the medial arm and axilla with cutaneous innervation.
5. **Medial Cutaneous Nerve of the Forearm (medial antebrachial cutaneous nerve)** – arises from the medial cord of the brachial plexus. Courses adjacent to the ulnar nerve to the superficial fascia of the forearm where it will branch to supply the anteromedial asp

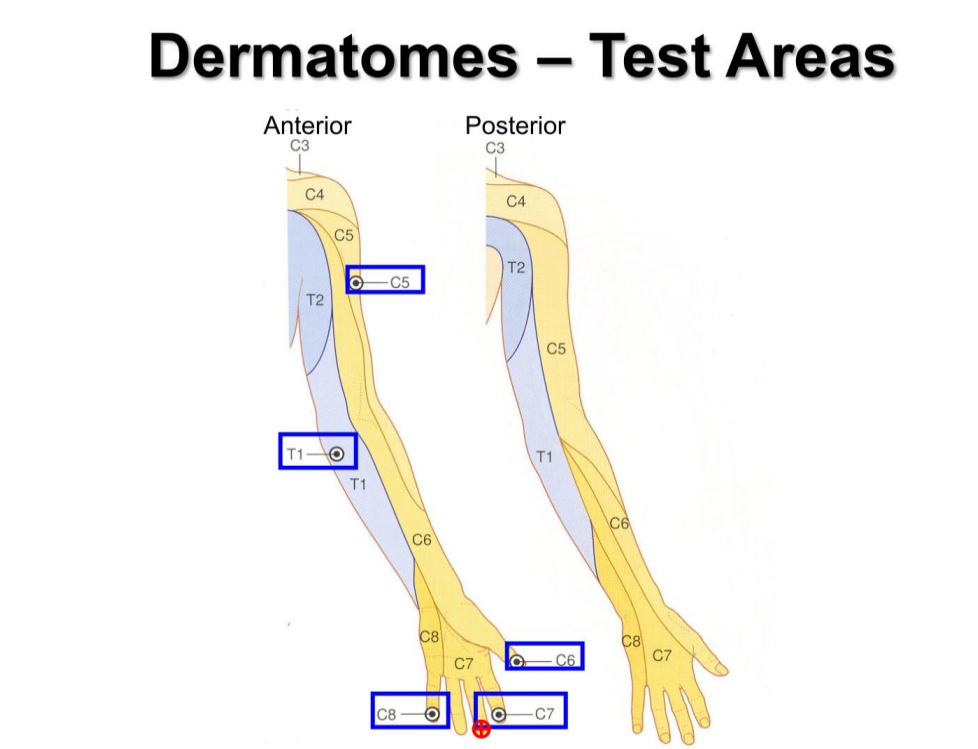


Radial Nerve (C5, C6, C7, C8, T1) – The radial nerve is one of the terminal branches from the posterior cord.

- It enters the arm posterior to the brachial artery, medial to the humerus and anterior to the long head of the triceps.
- It descends inferolaterally with the profunda brachii artery in the radial groove.
- Supplies the triceps brachii and then pierces the lateral intermuscular septum and continues anteriorly between the brachialis and the brachioradialis to reach the forearm.

Clinical Note: Due to the radial nerves close association with the humeral shaft as it runs within the radial groove it is susceptible to injury during humeral midshaft fractures. The radial nerve courses with the profunda brachii artery and thus it is also at risk of injury.





B. Cutaneous Innervation of the Upper Limb

1. **Posterior cutaneous nerve of the arm and forearm** (C5-C8) – branch of the radial nerve; supplies the skin on the posterior arm and forearm
2. **Lateral cutaneous nerve of the arm** (C6, C7) – branch of the musculocutaneous nerve; supplies the skin on the lateral forearm
3. **Medial cutaneous nerve of the arm** (C8-T2) – branch of the medial cord of the brachial plexus; supplies the skin on the medial arm
4. **Medial cutaneous nerve of the forearm** (C8, T1) – branch of the medial cord of the brachial plexus; supplies the skin on the medial and anterior forearm